Weather Forecast Simulation(C++)

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Problem Statement

Core Challenge

Build an interactive weather system managing multiple cities with real-time data analysis and user-friendly interface.

Primary Objectives

- Practice class-based design in C++
- Implement robust input validation
- Generate meaningful weather insights

Key Features

01

View Cities

Browse available locations

02

Check Weather

Single city detailed report

03

Compare Cities

Side-by-side analysis

04

Global Statistics

System-wide insights

```
1  ========================================
2          WEATHER FORECAST SIMULATION
3  ========================================
4   1. View Cities
5   2. Check Weather
6   3. Compare Two Cities
7   4. View Global Weather Statistics
8   5. Exit
9  ========================================
10
11 => Enter choice: 1
12
13 +----------------------------------------+
14 |              View Cities               |
15 +----------------------------------------+
16 |  1. Mumbai                             |
17 |  2. Delhi                              |
18 |  3. Bangalore                          |
19 |  4. Chennai                            |
20 |  5. Kolkata                            |
21 +----------------------------------------+
```

Sample Output

Table-Style Reports

Clear, organized console output with aligned columns for easy data reading

Comparison Format

Visual side-by-side analysis showing temperature and humidity differences

Statistical Insights

Average calculations, extreme value identification, and trend analysis

Concepts & Future Scope

Current Implementation

- **Encapsulation & Constructors**

  Data hiding principles and object initialization

- **Arrays of Objects**

  Collection management using C++ arrays

- **Control Flow**

  Loops, conditionals, and switch-case structures

- **Static Methods**

  Utility functions without instance creation

Future Enhancements

- **Persistent Storage**

  File or database integration for data persistence

- **Live API Integration**

  Real-time weather data from external services

- **Prediction Logic**

  Forecast algorithms using historical patterns

- **Modern Interface**

  GUI desktop or web-based frontend

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Thank You